S-PERFECTION OF A PSEUDO-COHERENT COMPLEX TESTED BY LOCAL BOUNDEDNESS ON THE FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12758-12770 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 12772-12778, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Fibre-testable perfection criterion; member of the chapter's relative-perfect toolkit with this wave's L16175/L16176.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma affine-locally-rel-perfect (etale-local character of relative perfection); Derived Categories of Schemes, lemma bounded-on-fibres (the affine criterion).

Source reference. Original lemma 0DKQ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12313–12335. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GEH: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12778–12778. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S such that the structure morphism $f : X \to S$ is flat and locally of finite presentation. Let E be a pseudo-coherent object of $D(\mathcal{O}_X)$. The following are equivalent*

(1) *E is S-perfect, and*
(2) *E is locally bounded below and for every point $s \in S$ the object $L(X_s \to X)^*E$ of $D(\mathcal{O}_{X_s})$ is locally bounded below.*

Proof. Since everything is local we immediately reduce to the case that X and S are affine, see Lemma 0DKQ. This case is handled by Derived Categories of Schemes, Lemma 0GEH. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.